Lake browning (DOC) and fish feeding ecology

Course: BIS15L

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Background

DOC or Dissolved Organic Carbon is when decaying plants or organisms causes the water to brown overtime. This makes it so the sunlight cannot penetrate through the water as easily and thus controls nutrition availability and water temperature. DOC reducing light penetration causes primary production is suppressed and reliance on algae-based food sources shifts to bacteria-based food sources

Data Source

-Charette et al. 2024 freshwater fish dataset - -
3,430 fish records, 78 lakes, 6 species, 6 regions
- Variables: DOC, isotopes, trophic position, pelagic diet proportion, fish length

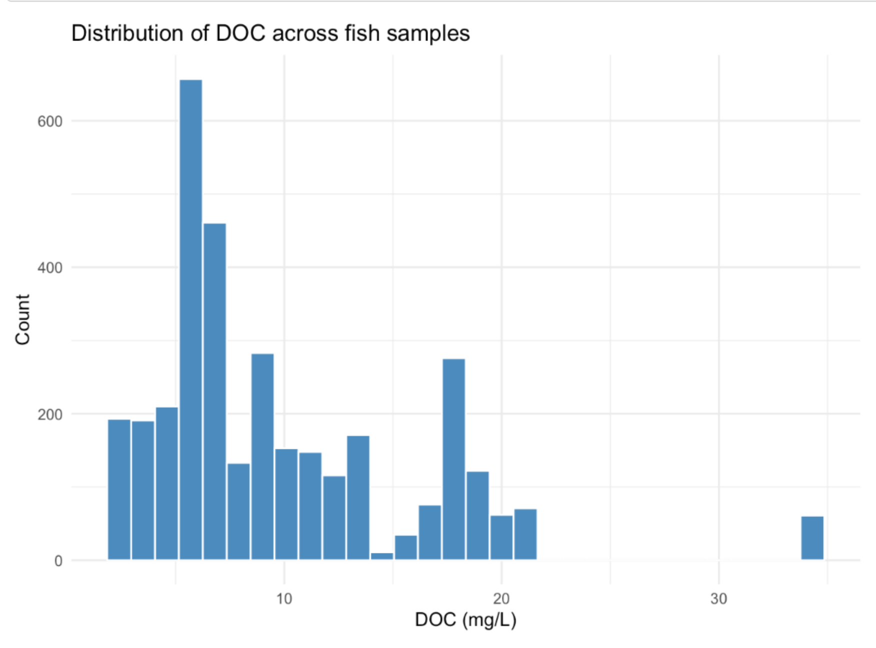
Research Questions

Question 1 : DOC vs trophic position
Question 2: DOC vs pelagic diet proportion
Question 3: Species/region differences in DOC relationships

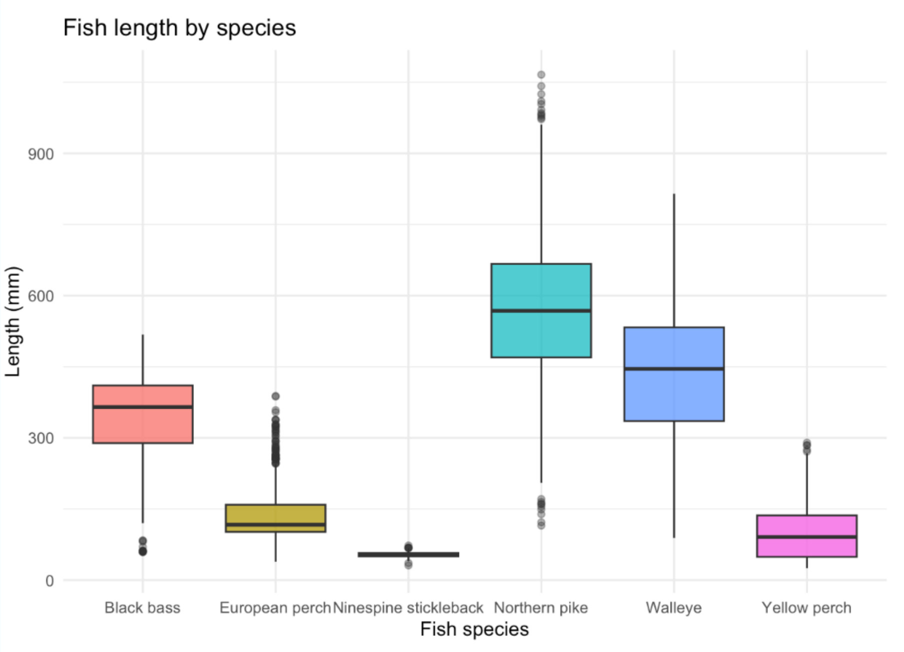
Workflow

- Import and clean two CSV files
- Join fish and baseline data by lake
- Create analysis dataset and DOC categories
- Check missingness and build summaries

Results

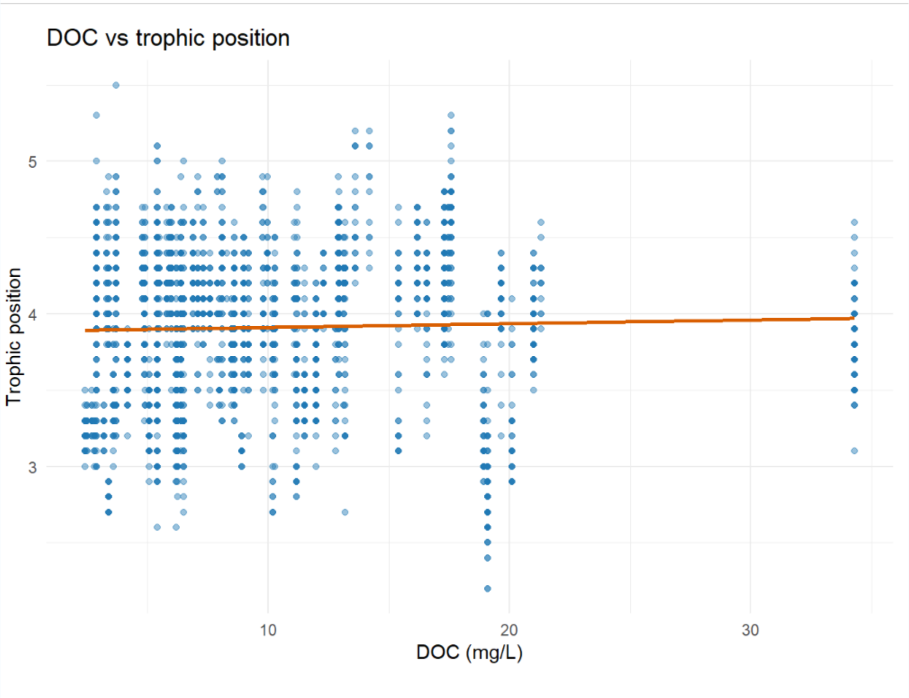


Distribution of DOC across fish samples
Count
DOC (mg/L)
0
200
400
600
10
20
30



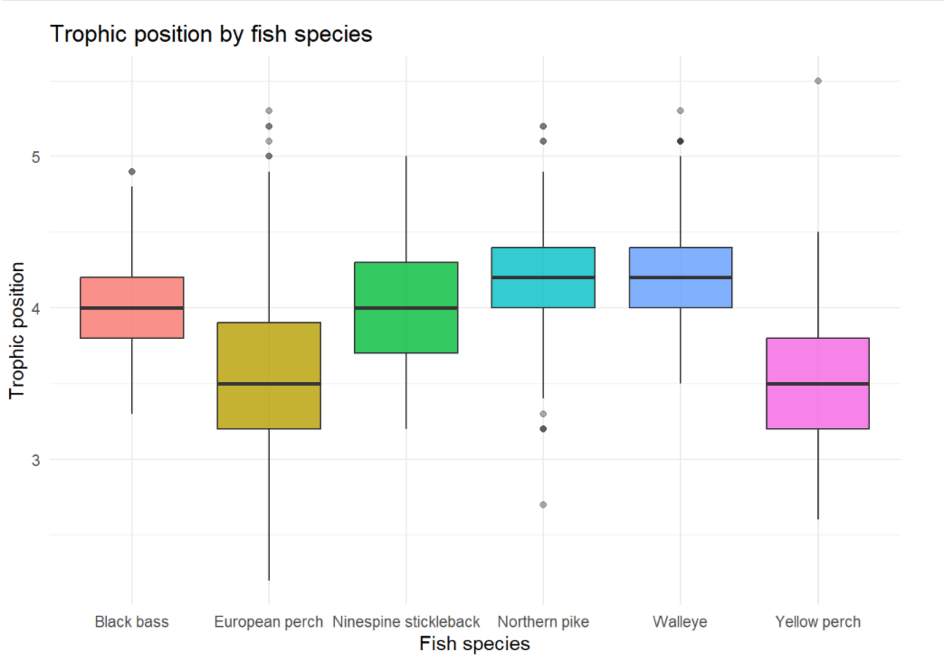
Fish length by species
Length (mm)
Fish species
0
300
600
900
Black bass
European perch
Ninespine stickleback
Northern pike
Walleye
Yellow perch

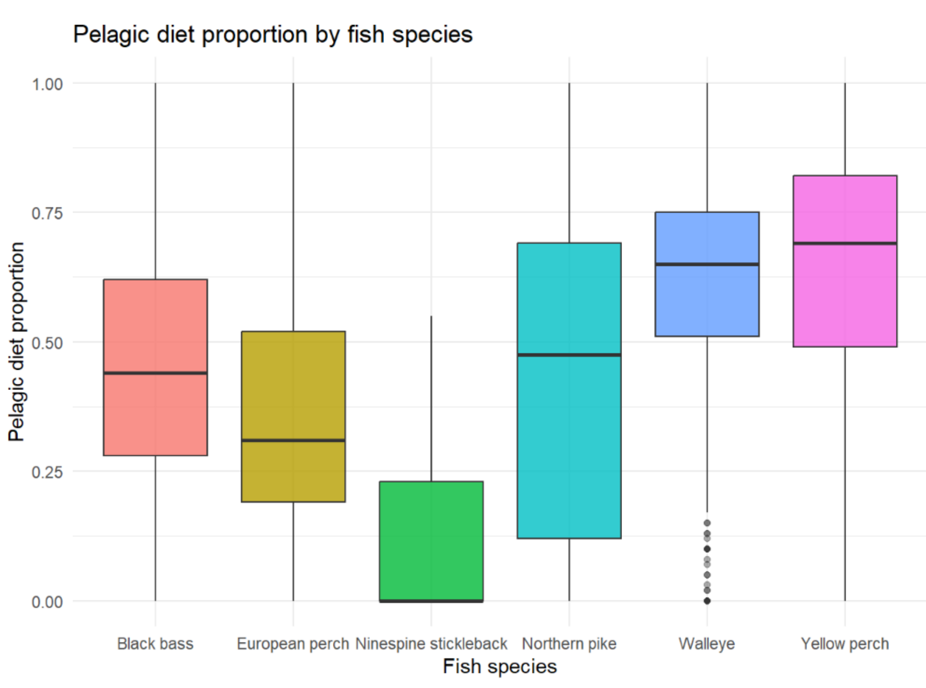
Does DOC relate to trophic position?



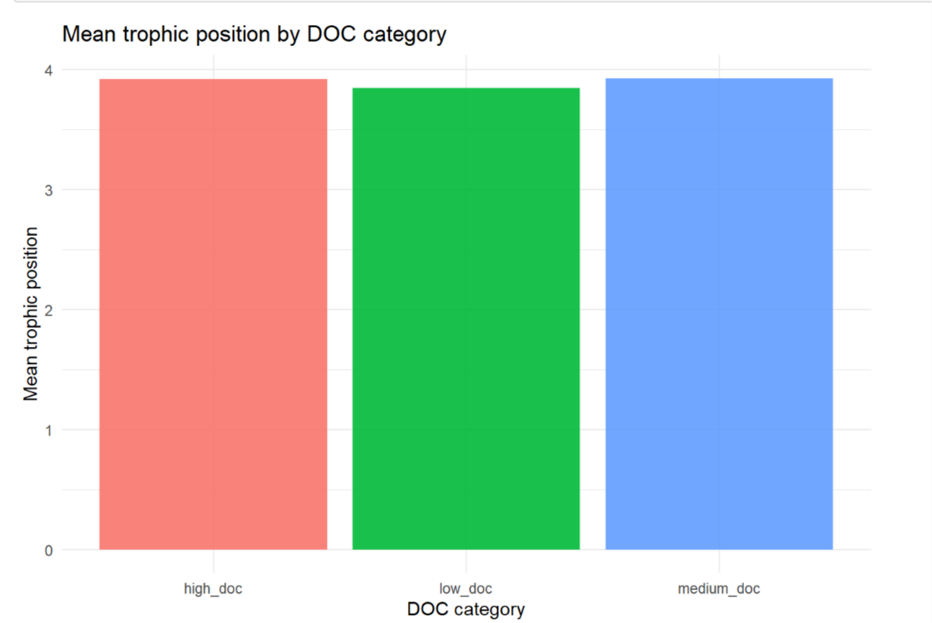
Does DOC relate to pelagic diet

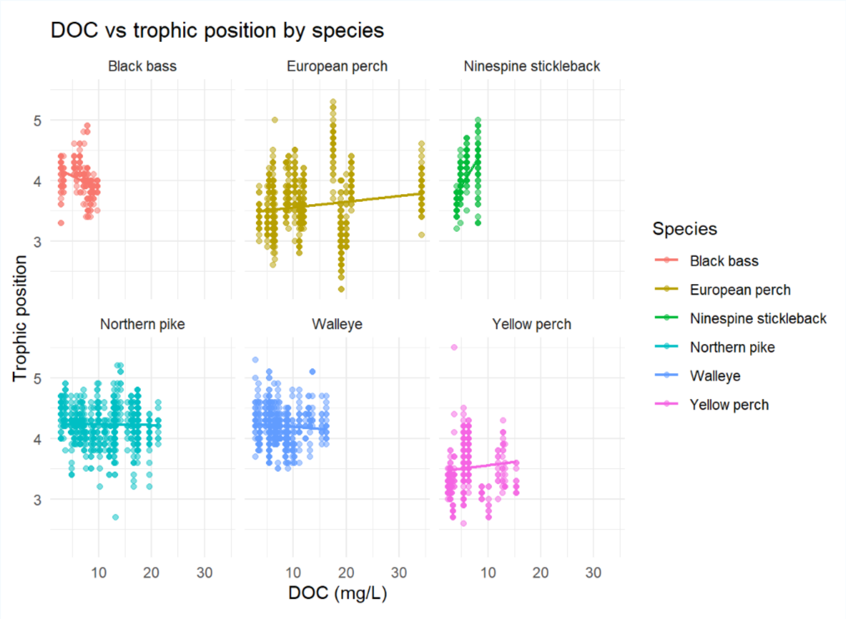
Do DOC patterns vary by species?



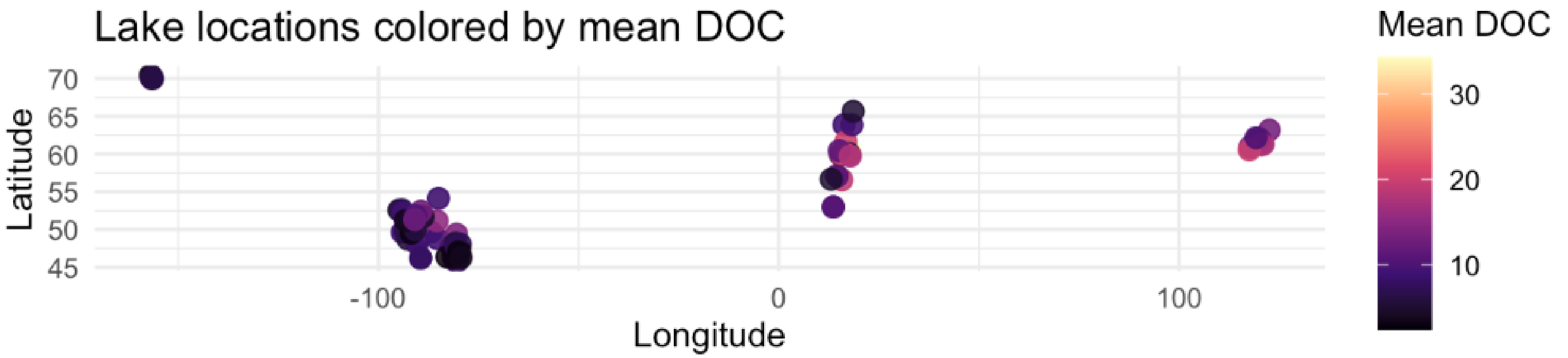


Do DOC patterns vary by species?

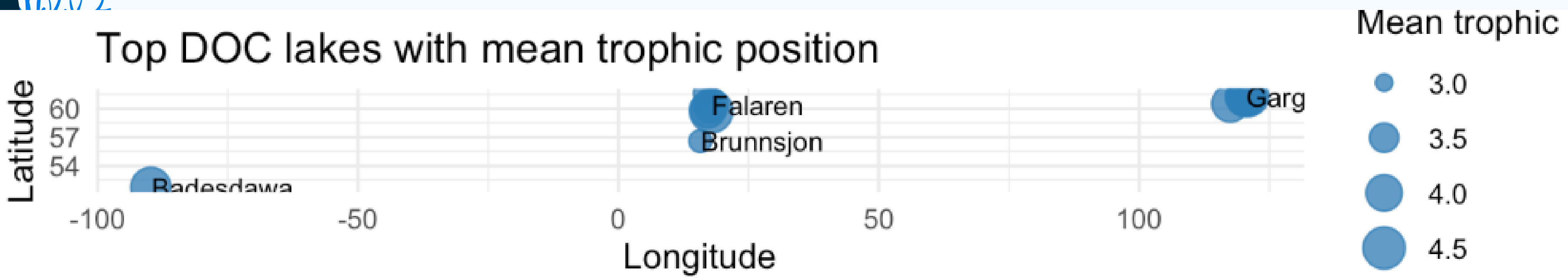




Mapping



Lake locations colored by mean DOC
Latitude
70
65
60
55
50
45
-100
0
100
Longitude
Mean DOC
30
20
10



Top DOC lakes with mean trophic position
Latitude
60
57
54
Falaren
Brunnsjon
Garg
-100
-50
0
50
100
Longitude
Mean trophic
3.0
3.5
4.0
4.5

Shiny App Demo

Limitations and Conclusion

- Overall, trophic position showed only a weak relationship with DOC.
- But pelagic diet proportion decreased as DOC increased.
- High DOC seem to shift **where** fish get their food from more than they shift **overall food-chain level.**
- Species differences are also a big reason the full-dataset patterns are not simple.